

Zheng Huang

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Education

Dartmouth College

PhD of Computer Science, GPA: 3.7/4, advisor: Prof. [Yujun Yan](#)

Sept 2023 - Present

Hanover, NH, United States

University of Virginia

Master of Computer Science, advisor: Prof. [Jundong Li](#)

Aug 2019 - Dec 2021

Charlottesville, VA, United States

University of British Columbia

UBC Visiting Student, Department of Computer Engineering

June 2018 - August 2018

Vancouver, BC, Canada

Hebei University of Technology

Bachelor of Computer Science, major GPA: 87/100

Sept 2015 - June 2019

Tianjin, China

Research Interest

I am interested in **Foundation Models** and **Machine Learning**, including:

- **Foundation Model Post-Training**: Studying supervised fine-tuning and inference-time methods for large language models and diffusion-based language models, with an emphasis on reasoning, uncertainty, and interpretability informed by neuroscience.
- **Graph Machine Learning**: Designing principled graph learning frameworks to improve generalization across heterogeneous graph data, with applications to recommender systems and real-world relational tasks.

Research Experience

Diffusion Language Model Inference with MCTS

June 2025 - Sept 2025

AWS AI Lab, Applied Scientist Intern — EACL, [Arxiv](#)

Santa Clara, CA, United States

- Investigated inference-time reasoning in diffusion language models by framing generation as a structured search problem.
- Developed an MCTS-based decoding framework to improve long-horizon reasoning under token-level uncertainty.
- Designed uncertainty-aware rewards and confidence-guided filtering to enable principled reasoning without additional training.
- Conducted extensive empirical and theoretical analyses of reasoning and sampling strategies in large language models.

Seeing Through the Brain: New Insights from Decoding Visual Stimuli

Sept 2024 - Jan 2025

Research Assistant, Dartmouth College, [Arxiv](#)

Hanover, NH, United States

- Proposed PRISM, a fMRI decoding framework that uses language models as a brain-aligned intermediate representation for diffusion-based image reconstruction.
- Supervised fine-tuned a language model on structured, object-centric text to align textual representations with fMRI signals.
- Designed an agentic LLM-based prompt optimization loop to discover brain-aligned object attributes and relationships.
- Introduced an object-centric diffusion framework with spatially grounded cross-attention for text-guided image generation.
- Conducted systematic analyses of supervised fine-tuning in large language models, and inference behavior in diffusion models.

Selected Publications

- **Zheng Huang**, Kiran Ramnath, Yueyan Chen, Aosong Feng, Sangmin Woo, Balasubramaniam Srinivasan, Zhichao Xu, Kang Zhou, Shuai Wang, Haibo Ding, Lin Lee Cheong. “Diffusion Language Model Inference with Monte Carlo Tree Search”, EACL 2026, [Arxiv](#).
- **Zheng Huang**, Enpei Zhang, Yinghao Cai, Weikang Qiu, Carl Yang, Elynn Chen, Xiang Zhang, Rex Ying, Dawei Zhou, Yujun Yan. “Seeing Through the Brain: New Insights from Decoding Visual Stimuli with fMRI”, Under Review, [Arxiv](#).
- Weikang Qiu, **Zheng Huang**, Haoyu Hu, Aosong Feng, Yujun Yan, Rex Ying. “MindLLM: A Subject-Agnostic and Versatile Model for fMRI-to-Text Decoding”, ICML 2025, [Arxiv](#).
- **Zheng Huang**, Qihui Yang, Dawei Zhou, Yujun Yan. “Enhancing Size Generalization in Graph Neural Networks through Disentangled Representation Learning”, ICML 2024, [Arxiv](#).
- **Zheng Huang**, Jing Ma, Yushun Dong, Natasha Zhang Foutz and Jundong Li, “Empowering Next POI Recommendation with Multi-Relational Modeling”, SIGIR 2022. [Arxiv](#).
- [Full Publication List \(Google Scholar\)](#)

Industry Experience

Alexa Speech Recognition, Amazon.com, Inc.

Mar 2022 - Jan 2023

Machine Learning Engineer

Seattle, WA, United States

- Developed Federated Learning (FL) systems to preserve users’ privacy and improve the quality of speech recognition.
- Worked on a team and delivered an on-device FL Recurrent Neural Network Transducer prototype that is capable of learning from the audio without relying on sending users voice recordings to the cloud.
- Implemented FL on-device trainer that was constructed during the device idle time and deconstructed after the completion of training tasks to minimize the footprint of training in memory to avoid customer friction.
- Conducted in-depth investigation on decentralized machine learning, privacy-preserving and AWS infrastructure.

Technical Skills

- **Programming:** Python, Java, C++, C, R, Javascript, HTML, CSS, SQL, MATLAB.
- **Tools:** PyTorch, TensorFlow, Scikit-learn, PySpark, AWS, Linux, Numpy, Pandas, Latex.

Services

- **Conference reviewer:** ICLR 2025, NeurIPS 2025, ICLR 2024, NeurIPS 2024, ICLR 2023, NeurIPS 2024, ICLR 2023, JMLR 2022, ECML 2022, PAKDD 2021, WSDM 2021
- **Industry reviewer:** Amazon Machine Learning Conference 2022

Awards

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| • ICML Travel Grant | 2024 |
| • SIGIR Student Grant | 2022 |
| • Compuster Science Department 2nd-level Academic Fellowship | 2018 |
| • Compuster Science Department 1st-level Academic Fellowship | 2017 |
| • Compuster Science Department 2nd-level Academic Fellowship | 2016 |

Teaching

- Teaching Assistant, Deep Learning, Fall 2025
- Teaching Assistant, Deep Learning, Spring 2025